



Determinants of innovative green electronics: An experimental study of eco-friendly laptop computers

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ABSTRACT

Considering the nature and complexity of electrical/electronic product materials, their energy consumption, and the e-waste they create, sustainability should be included in their design and innovation. This study examines the benefits of eco-friendly laptop computer innovations that allow users to leverage multiple features to co-develop green electronics products that achieve premium performance. Using conjoint analysis, the authors identify the relative importance of various attributes of laptops, as perceived by users. The most important attribute of laptops is price, followed by battery, shell, central processing unit (CPU), monitor, storage device, and keyboard. The most preferred combination of attributes for an eco-friendly laptop consists of median price, solar battery, alloy-based shell, multicore CPU, LED monitor, SSD storage device, and recycled paper keyboard. Various standards, such as energy savings, waste recycling, pollution prevention, and compatibility, can be derived from these attributes to develop new functions and modules. By reflecting on the utilities of certain attributes, computer manufacturers can increase user value, improve waste management, reduce energy consumption, and advance their electronic product development efforts.

1. Introduction

In the past decade, technology and product development in the electronics industry has influenced society, markets, and the environment. The production of electrical/electronic equipment is one of the fastest growing manufacturing domains in the world (EC, 2000). However, in 2014, the quantity of waste electrical/electronic equipment that had reached its end-of-life condition was 41.8 million tons, with 50 million tons of waste expected before 2019 (Baldé et al., 2015). Such waste consists of complex mixtures of components that contain hazardous substances. Its significant environmental impact spans all stages of the product life cycle, including extraction and use of raw materials, consumption of energy and other resources by production and product use, and waste management (Dumitrescu et al., 2014). In response to the rise of international environmental regulations such as the Montreal Protocol, European Union environmental directives (Waste Electrical and Electronic Equipment [WEEE], Restriction of Hazardous Substance [RoHS], energy-using products [EuP]), and the Kyoto Protocol, companies in the electronics and consumer products industries—such as LG, Acer, Samsung, Vivo, and Dell—have begun to design “green” electronics that treat environmental attributes as design objectives rather

than constraints. This emerging business innovation and sustainability in electronic markets is being intertwined with environmentally considerate design and manufacturing to increase the number of eco-friendly electronic components. However, production activity goes beyond the application of technologies to realize useful energy transfers; it also involves understanding users’ willingness to pay for such consumption value. Although companies should be prepared to minimize the environmental impacts of construction components regardless of users’ awareness of those impacts, our objective is to determine which attributes of green electronics laptops users value most. We also seek to examine how changes in these attributes—resulting from environmental and innovative standards—affect eco-friendly laptop design and development.

Electronics deliver higher efficiency, greater flexibility, and added features to many electrical appliances. Increased consideration of environmental sustainability has led to a more flexible and relational framework to develop and change the nature of electronics. Existing research has addressed several issues, such as policy and regulation (Awasthi and Li, 2017; Ciocci and Pecht, 2006; Thompson, 2019), end-of-life processes (Kaewunruen et al., 2019; Sun et al., 2015), provision of information systems (Gu et al., 2017; Wang and Wang, 2017),

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materials and energy utilization (Cho et al., 2015; Gao et al., 2018; Mauro et al., 2019), and supply chain management (Ayvaz et al., 2015; Shimada and Van Wassenhove, 2019). Such research establishes both sustainability-oriented models and in-depth theoretical understanding of electronics product development.

Despite the multiplicity of influences and ongoing debate about the desirability of development versus the imperative of sustainability, we can identify some key prerequisites of product success. Innovation is critical for increasing the environmental benefits of new lead-free electronic products (green electronics) (Mueller et al., 2005). Such innovation includes making significant, demonstrable progress toward improving environmental performance by reducing environmental impacts and achieving efficient, responsible uses of natural resources (Díaz-García et al., 2015). Firms are exploring new means of designing environmental technologies and products to achieve sustainability, reduce waste, and demonstrate social responsibility. Several studies (Chang, 2019; Chen, 2014; Jabbour et al., 2015; Jannat, 2016; Ma et al., 2017; Porter and van der Linde, 1995) indicate that companies that engage in green innovation not only minimize production waste and cost but also drive value creation, enhance technical knowledge, and increase competitiveness; they satisfy market demand and meet increasingly strict international environmental protection regulations. Successful *green innovation* is a driving force that offers greater efficiency to firms and enables them to create and incorporate changes in products and production processes that contribute to firm profitability.

Any type of innovation-design planning and practice must satisfy consumer needs. Without consumer acceptance, even highly organized and efficient production systems will collapse (Fletcher and Goggin, 2001). Notably, consumer environmentalism is increasing; consumers support better, efficient uses of resources. Therefore, firms must learn how to incorporate the dynamics of both consumers and technologies into their product development. They must review and reconstitute their products and services. Fortunately, such creativity- and collaboration-oriented approaches can stimulate new trains of thought, establish new connections, and lead to new ideas for involving users in green innovation. Consumers can take proactive roles in defining and negotiating their own product needs to meet their explicit and implicit requirements. Accordingly, the collaboration of users in the product development process is relevant to enhancing environmental value. Such collaborative design not only delivers core benefits to users but also improves eco-performance and manages any necessary trade-offs with existing core/auxiliary product benefits.

Several tools help guide sustainable product development; they include environmentally conscious quality function deployment (ECQFD) (Vinodh and Rathod, 2010), life cycle assessment (LCA) (Curran, 2013; Finnveden et al., 2009), design for environment (DFE) (DeMendonça and Baxter, 2001), environmental impact assessment (EIA) (Jay et al., 2007), and green procurement (Blome et al., 2014). Although these tools focus on the environmental effects of products, they perform relatively poorly in terms of translating user needs and requirements into technical specifications. For producers, developing valuable eco-friendly attributes is a decision that requires consideration of responders (Alriksson and Öberg, 2008). A common problem with measuring one single attribute at a time is that respondents are subject to a ceiling effect. The conjoint analysis (CA) gives an advanced approach to solve this problem (Lawson et al., 2006). The development of green electronics products according to sustainability specifications reveals the need for CA to estimate the structures of users' preferences, given their overall evaluations of sets of alternatives that are pre-specified in terms of attribute levels. According to CA, every combination of attributes is a product idea, and the tester is the stimulus; together, they provoke users' overall assessment values for a certain stimulus. Green electronics producers generate innovative ideas and put them into practice in the shortest possible time; customers base their evaluations on the proffered combinations of product attributes and choose eco-efficiencies that provide them with maximum sustainable

benefits. In the following section, we review prior literature to contextualize the state of innovation and fashion. We also recommend approaches for understanding green innovation (which we interchangeably refer to as environmental innovation) and new product development (NPD). After outlining our data collection methods and describing our analysis results, we conclude with a discussion of our main findings.

2. Literature review

2.1. Innovation and fashion

In general terms, innovation reflects embodiments of marketable new ideas (Fitzgerald et al., 2011; Vyas and Woodside, 1984) that can address real-world problems. Therefore, innovative ideas should reflect feasible possibilities, such that they can actually be implemented in relevant contexts. According to Roberts (1988: 12), "innovation is composed of two parts: (1) the generation of an idea or invention, and (2) the conversion of that invention into a business or other useful application". Furthermore, various elements that constitute an innovation are continuously put into balance, in an effort to conceive of their potential contributions to the process of transforming an idea into a real innovation. The pathway generally starts with basic scientific discoveries, leads to improved products and greater quality, and finally results in the commercialization and distribution of the innovative offering to customers through markets (Chesbrough, 2006). Each step requires knowledge, starting with science, technology, and development insights but also including knowledge of basic or applied problems that need a solution (Freeman, 1982). For example, Paper Polypropylene (PP) Alloy is a paper-based composite material, made of recycled paper and polypropylene, that has been used successfully to create shells for consumer electronics (Designer, 2011). The material has comparable strength and elasticity to acrylonitrile-butadiene-styrene (ABS) plastics, can be recycled, and is low in cost (IF Design, 2021). These material properties, combined with satisfactory mechanical condition considerations, provide design flexibility to create truly environmentally friendly and long-lasting laptop components.

Beyond these knowledge considerations, the vast investments required by innovative activity requires the firm to prioritize and manage its projects according to their commercial potential. Some companies seek immediate financial gains and focus on generating revenues rather than developing new ideas or quality connections for the long term. In this sense, fashion, defined as "the pursuit of novelty for its own sake" (Robinson, 1958: 127), offers a relevant conceptual perspective, in that entrepreneurs might attempt to be fashionable to attract publicity and generate sales. Regardless of whether a product offers improved functionality, if a consumer has never seen it before, it is novel (Liao, 2019), which evokes a sense of newness, regardless of any objective standard or definition of a true innovation.

Fashion also requires "the imitation of a given example [that] satisfies the demand for social adaptation" (Simmel, 1957: 543), in a setting in which people are aware of and influenced by other people's choices. Using similar elements to form different styles is compelling, assuming those elements represent the current fashion. Through processes of reinvention in cycles, novelties appear at regular intervals, such that in the presence of a limited set of choices, novelty entails the rediscovery of old styles. Even if a new design stems from a dedicated, creative process, because fashion does not really produce different contributions, consumers might not view a rediscovery of an old idea as an innovation (Kadirov, 2015), especially if they rely on their own principles and quality standards to make consumption decisions. Such reasoning implies that firms that pursue success in the form of immediate revenue gains might fail to exploit the full potential of their innovations and inhibit structural progress, from a wider view.

2.2. Green innovation and new product development

In the sustainable development era, firms should integrate ideas for protecting the environment into their product offerings. A key feature of sustainability is that adopted solutions are flexible enough to allow new scientific findings, engineering solutions, and innovations to be incorporated into existing systems (Andersson, 2004). In the course of rethinking corporate sustainability, researchers widely acknowledge that innovation is of utmost importance for achieving sustainability (Hansen et al., 2009). Efficient, strategic implementation can help firms develop innovation portfolios, enhance environmental performance, and increase value, resulting in greater profitability.

The concept of green innovation revolves around new or improved practices, processes, techniques, systems, and products to prevent or minimize environmental damages. It is closely related to notions of environmental innovation, sustainable innovation, and ecological innovation, all of which authors refer to interchangeably (Schiederig et al., 2012). Chen et al. (2006: 332) define green innovations as “hardware or software innovation that is related to green products or processes, including innovation in technologies that are involved in energy saving, pollution prevention, waste recycling, green product design, or corporate environmental management.” Several studies (Driessen and Hillebrand, 2002; Mardani et al., 2017; Schiederig et al., 2012) confirm that green innovation refers to creative ideas and environmental initiatives, products, services, and processes that reduce negative environmental impacts and/or yield environmental benefits, while creating value in the market and achieving product differentiation.

Green innovation aims to adjust the design of existing products or processes to increase abilities and achieve sustainability by satisfying stakeholders and regulatory agencies; it leads to new business models, changes competitive rules, and generates business opportunities. Therefore, green innovation is a win-win solution to the conflict between economic development and environmental protection. However, some firms claim green innovation inaccurately, seeking to achieve sales of relatively poor quality products, compared with traditional products. For example, they might promote commodities that become prematurely obsolete, to accelerate repurchases. Such a strategy relies on notions of fashion, in which any novel design can be appealing, as long as it is new and establishes some claim to exclusiveness (Gronow, 2009), even if that design has no other function.

In addition to fashion perspectives, authors have proposed a variety of theories to describe the implementation of green innovation. For example, stakeholder theory suggests that as a response to stakeholders' environmental demands, firms should consider innovation strategies that account for stakeholder groups' reactions to salient environmental practices and business performance (Geffen and Rothenberg, 2000; Hsu et al., 2013). According to the resource-based view (RBV), firms can distinguish themselves from their competitors by using available resources in innovative ways to avoid pollution, reduce waste, save energy, and improve the design and quality of their products (Berchicci and Bodewes, 2005; Chang and Chen, 2013; Chen et al., 2006; Lin and Chen, 2016). Institutional theory posits that green innovation implementation represents firms' responses to institutional pressures and configurations for environmental protection (Aguilera-Caracul and Ortiz-de-Mandojana, 2013; Liu et al., 2019). Social capital theory accounts for firms' abilities to acquire resources through relationships embedded in social units that provide support for their innovations (Chen and Hung, 2014), and contingency theory proposes that firms face rapidly changing market conditions and undertake green innovations in response to the pressures of environmental regulations and environmental dynamism (Dangelico, 2016; Tsai and Liao, 2017).

Extensive literature on green product innovation focuses on the organizational aspects of green innovation, including the antecedents of green product innovation activities (Chen et al., 2012), the effects of various contexts on green product innovation (Chan et al., 2016; Liao,

2019), and links between green performance and financial performance (e.g., Huang and Wu, 2010; Zailani et al., 2015). Although these studies provide important insights, they do not offer sufficient information about what companies should do to integrate environmental sustainability into NPD. Dangelico and Pujari (2010) suggest that gaining more insight into green NPD (GNPD) may be a starting point for improving this process and lowering the failure rate of green product innovations: GNPD explicitly integrates environmental issues to create less environmentally harmful products. It is a broad concept that encompasses a range of issues, from redesigning existing products to creating new products and services driven by environmental concerns (Berchicci and Bodewes, 2005). It is characterized by changes in products' sustainability characteristics during the development phase.

Because of the importance of environmentally responsible NPD, various concepts have emerged to support GNPD outcomes (Ellram et al., 2008). High levels of stakeholder integration are required to sustainably develop products; according to Li et al. (2019), corporate stakeholders have a significant impact on green manufacturing. Sharma and Henriques (2005) also find that stakeholders influence sustainability practices. Stakeholder orientation has implications for the GNPD community at large; the community benefits from more structural consideration of stakeholders during the decision-making phase, because stakeholders may possess knowledge and know-how that anticipates potential solutions and identifies innovative opportunities and future business models. Among stakeholders, users are important exchange partners, because they are willing to work toward sustainable outcomes (Sancha et al., 2016). Both extant literature (Joshi and Rahman, 2016; Thongmak, 2016) and common parlance acknowledge the existence of powerful green users who can influence environmental responsiveness. Thus, green product manufacturers gain more valuable information from user involvement during the GNPD process. User-centered solutions provide alternative strategies for changing sustainable products and features to maximize performance; they increase utilities and unlock benefits. This alteration of the traditional model of product production and development satisfies the logic of user needs; it focuses on the value of users to product development and uses their capabilities and characteristics to benefit from user involvement (Vermeulen et al., 2014). Such hands-on, iterative, collaborative development activities enable organizations to learn about and respond to real customer needs; they enhance product effectiveness, improve human well-being, and increase sustainability.

2.3. Green electronics design and development

The new field of green electronics plays a key role in the efficient, effective use of electricity. Many firms require particular types of products, such as laptop computers, cell phones, televisions, and washing machines, to incorporate lifecycle-based environmental considerations into product design, material sourcing and selection, manufacturing processes, and end-of-life product management. Most electronic products promote reuse and recycling by imposing collection and recovery quotas and reduce e-waste by enhancing product design. Green electronics must be designed in ways that reuse and recycle end-of-life products, with all components and materials considered and facilitated. Specific features, reflected in product design, provide significant feedback on mechanisms that enable electrical and electronic equipment to be reused. This notion provides the basis for our proposed approach to integrating environmental features throughout the NPD process, with particular focus on sustainable consumption and production.

Electronics products such as laptops contain hazardous substances and waste energy. Hazardous components commonly found in laptop devices include lead, mercury, cadmium, and brominated flame retardants (BFRs) in plastics. If disposed of carelessly, these toxic materials can be harmful to both people and the environment. With regard to waste energy, the power consumption of laptops varies with the

component; monitors and central processing units (CPUs) consume the largest share of power (Kumar and Kannegala, 2012; Maesri and Varadhan, 2005). Monitors have either liquid crystal displays (LCDs) or light-emitting diodes (LEDs); of the two, the latter consumes less power. Although LCD monitors can be energy-efficient (energy consumption of the backlight is the main electrical draw), reducing power consumption by about 30% (DisplaySearch, 2011), LED monitors are more efficient than LCD monitors by about 10%–30% (Park et al., 2013). The more intensive the operations processors run, the greater the energy their CPUs consume, such that advanced multicore processors consume relatively less energy for the same amount of processing done by multiple single-core processors (Boger and Dusthimer, 2013). Hard disk drives (HDDs) also are major laptop power users, making energy-efficient storage devices (e.g., solid-state drives [SSDs]) more important. According to Al Assaf et al. (2013), HDDs consume 3.7 W on read/writes when in active mode and 0.5 W when in standby mode, whereas SSDs consume 150 mW when active and 100 mW when idle.

Laptop batteries generally have short lifetimes; they need to be replaced a few times in a laptop's life, thereby contributing to e-waste. Solar batteries are renewable energy sources, in that their integrated energy conversion and storage systems benefit from efficient and economic storage and minimize energy losses during discharging (Luo et al., 2017). Nickel-metal hydride (NiMH) batteries are the dominant form of energy in portable applications because of their high energy density and low environmental hazard; they are more than 40% more energy-effective than nickel-cadmium (NiCd) batteries (Goswami and Kreith, 2016). Laptop shells and keyboards can be made from recycled paper, recycled plastic, or magnesium/aluminum alloy forms. Producing recycled paper creates 35% less water pollution and 74% less air pollution than producing virgin paper (Tabassum et al., 2017). Industry quotes for typical energy savings from producing recycled paper range from about 28% to 70% (Virtanen and Nilsson, 1993). Recycled plastic is abundantly available in the world as a waste material. Using recycled plastic to manufacture new plastic products can reduce the energy needed for production by about 75% (Laville and Taylor, 2017); recycled plastic is far cheaper than virgin plastic and has a more positive environmental impact, thereby justifying its use. Recycling high magnesium/aluminum alloys reduces waste, saves energy, conserves natural resources, lessens the use of municipal landfills, and provides recyclers and municipalities with considerable revenue. Finally, price is a key decision-making criterion for purchasers of green products (Bhatia and Jain, 2013; Esmaili and Fazeli, 2015). Pricing must account for costs, profit targets, competitor price levels, and users' willingness to pay. The user element is complex: The business must sell products to users, and users' evaluations of products determine the prices they are willing to pay and reflect their perceptions of value. The price of any product is a function of its utility-bearing attributes or characteristics, as assessed by users (Rosen, 1974), who consider product prices to be valuable reference points. Utility information indicates varying cost performance among products, and users commonly use it as a criterion to choose products.

3. Methodology

The CA method is one of the most widely used quantitative tools for helping firms determine structural designs to achieve different combinations of multiple-attribute product standards. It uses customers' preference estimations for sets of experimental product concepts to identify "optimal" designs to be tested. The use of CA to assess environmental features has gained increasing attention, because it can determine explicit trade-offs in environmental conditions by discovering preference structures as well as monetary valuations (Farber and Griner, 2000). The use of CA allows us to determine the optimal features of green products (Alriksson and Öberg, 2008; Sonnenberg et al., 2014), assess which products consumers will choose, and estimate the weights

customers assign to various factors that underlie their decisions. Alternative methods to measure consumer preferences in conjoint experiments are as follows: part-worth, ideal vector, and ideal point models. A more detailed discussion on these models is presented by Green et al. (2001) and Gustafsson et al. (2003). In practice, conjoint studies use the part-worth model because of its flexibility and ability to estimate the greatest number of parameters. The model reflects respondents' evaluative judgments and predicts how consumers will react to changes in attributes of new products (Chayet et al., 2004). It assumes that all interaction terms (attribute levels) are negligible, and it has a simplified structure. The model reflects a utility function over subsets of attributes, estimating part-worth utilities of each attribute level using ordinary least squares (OLS) (Rao, 2010). The analysis decomposes the preferences structure to derive the part-worths (regression coefficients) for each attribute level. For a data regression involving known data (e.g., " Z_{nk} ", " x_{kij} ") with estimated parameters (e.g., " λ_{nij} "), the part-worth model is:

$$Z_{nk} = \sum_{i=1}^I \sum_{j=1}^{J_i} \lambda_{nij} x_{kij} + \varepsilon_{nk}, n = 1, \dots, N, k = 1, \dots, K \quad (1)$$

where Z_{nk} = the preference judgment of respondent n for stimulus profile k ; λ_{nij} = respondent n 's part-worths associated with the j -th level ($j = 1, 2, \dots, J_i$) of the i -th attribute; x_{kij} = dummy variable representing the presence ($x_{kij} = 1$) or absence ($x_{kij} = 0$) of the j -th level of i -th attribute in the k -th profile; and ε_{nk} = error terms. To encourage elucidation, we can rescale the λ_{nij} s acquired from the regression, such that the smallest preferred level of each attribute is set to 0 and the largest preferred profile combination is set to 100, creating results that can be effectively translated and understood. Using λ_{nij} s to denote the estimated part-worth, the utility u_{nk} of a profile k to respondent n is equal to

$$u_{nk} = \sum_{i=1}^I \sum_{j=1}^{J_i} \lambda_{nij} x_{kij} \quad (2)$$

We also can use the utilities to determine the most preferred attribute levels of the product or profile. We scale them to sum to 0 within each attribute (i.e., they are zero-centered), such that we can compare differences between attribute levels. Because all utilities are expressed in a common unit, we can calculate importance values by taking the utility range for each attribute separately and dividing it by the sum of the utility ranges for all attributes. The importance scores can determine the attribute processing strategies adopted by a respondent (Hensher, 2006; Hensher et al., 2012). We then calculate the relative attribute importance (RAI) for each attribute:

$$RAI_{ni} = \frac{\max\{\lambda_{ni1}, \lambda_{ni2}, \dots, \lambda_{niJ_i}\} - \min\{\lambda_{ni1}, \lambda_{ni2}, \dots, \lambda_{niJ_i}\}}{\sum_{i=1}^I (\max\{\lambda_{ni1}, \lambda_{ni2}, \dots, \lambda_{niJ_i}\} - \min\{\lambda_{ni1}, \lambda_{ni2}, \dots, \lambda_{niJ_i}\})}, n = 1, \dots, N, i = 1, \dots, I \quad (3)$$

3.1. Formulating attributes and levels

Defining proper attributes and levels is arguably the most fundamental step in designing a good conjoint study. With regard to the selection of eco-friendly laptop attributes and levels, we considered several principles. First, focus group interviews with users indicated which accessories they considered when choosing laptop quality. The focus groups provided in-depth, qualitative information about topics of NPD interest, related to identifying the appropriate attribute level. From two interviews of different consumer groups, featuring 24 users in total, we identified 18 primary attributes. Because the resulting analysis would be complex, we redefined the attributes according to the mean and operational considerations. We excluded 7 attributes that were closely related and 4 attributes that were inconsistent with aspects connected to eco-friendly laptop design.

Second, in interviews with professional designers, each of whom had significant computer design experience and worked for computer brands, we learned about current trends in eco-friendly laptop accessories: monitors (LED and LCD), CPUs (multicore and multiple single-core processors), storage devices (HDD and SSD), batteries (solar, NiMH, and NiCd), shells (recycled paper, recycled plastic, and magnesium/aluminum alloys), keyboards (recycled paper, recycled plastic), and price (28,000 New Taiwan [NT] dollars, 38,000 NT, and 48,000 NT; 1 USD = 31.5 NT). These attributes and their levels represent the most relevant features to characterize a new product that would achieve needed improvements (Table 1). To evaluate the seven eco-friendly laptop attributes that users most value (i.e., degree to which new products are consistent with customer values and practices), we used innovation evaluations related to energy savings, pollution prevention, waste recycling, and compatibility (Chen et al., 2006; Rogers, 2003). These evaluations allowed us to construct a data matrix that represented customer perceptions of each alternative on each of the observed attributes, with response values as integers ranging from 1 (“not preferred”) to 7 (“greatly preferred”) for each attribute. We used these attribute levels to construct or identify various stimuli profiles.

3.2. Stimuli set construction

We presented a stimuli set to respondents for the evaluation task, using critical eco-friendly laptop attributes, in an experimental design. Researchers commonly use a full-profile approach, which requires respondents to evaluate each profile holistically and in the context of all other stimuli. With seven attributes, four at two levels and three at three levels ($2^4 \times 3^3$), we construct a total of 432 ($2 \times 2 \times 2 \times 2 \times 3 \times 3 \times 3$) profiles. To reduce respondents’ potential information overload, we selected a fractional factorial design and constructed a set of 36 profiles using orthogonal array design (see Fig. 1). We asked respondents to provide preference ratings for each profile in the estimation set, on a continuous scale from 1 (“least preferred”) to 10 (“most preferred”).

3.3. Data collection

We obtained samples through purposive sampling, a method that involves an effective selection of sampling units within a segment of the population that is likely to yield samples that will offer the best estimate of a population parameter of interest (Guarte and Barrios, 2006). That is, the researcher decides what needs to be known and sets out select information-rich cases, to maximize the utilization of available resources (Patton, 2002). Therefore, the sample frame consisted of electronics product consumers at ten retail stores in Taipei. The respondents had already purchased eco-friendly laptops and electronics equipment.

Table 1
Attributes and levels for conjoint analysis.

Attributes	Levels
1. Monitor	LED LCD
2. CPU	Multicore processors Multiple single-core processors
3. Storage device	HDD SSD
4. Battery	Solar battery NiMH battery NiCd battery
5. Shell	Recycled paper Recycled plastic Magnesium/aluminum alloys
6. Keyboard	Recycled plastic Recycled paper
7. Price	28,000 NT 38,000 NT 48,000 NT

Of the 380 questionnaires distributed, 326 were returned; we eliminated 9 invalid questionnaires, leaving 317 valid questionnaires, corresponding to a high retrieval rate of 85.79% and a valid return rate of 83.42%. We measured reliability using Cronbach’s α coefficient, calculated for all items, to assess the internal consistency of the model variables. According to Nunnally (1978), a standard coefficient α of 0.70 or higher is acceptable. In our case, the value of the Cronbach’s α for the observed stimuli was 0.89, indicating that our study achieved good consistency. The conjoint procedure also computes correlations between the observed and predicted rank order for the holdout profiles as a check on the validity of the utilities. The predictive validity of the Pearson correlation showed that the conjoint model offered good predictive ability ($r = 0.73$, $p < 0.01$). (The fitness and efficiency of a conjoint model indicates its ability to replicate reality, thus signaling support for validity.)

4. Analysis results

4.1. Descriptive statistics

Of the respondents, 48.6% ($n = 154$) were men, and 51.4% ($n = 163$) were women; 53.3% had monthly incomes of 25,001 NT to 45,000 NT. Ages ranged from 20 to 75 years, with 21.8% categorized as young (≤ 29 years), 68.1% as middle-aged (30–49 years), and 10.1% as older (≥ 50 years). More than one in two (50.5%) possessed higher levels of education (post-secondary school diploma or degree). Table 2 contains the sample characteristics.

4.2. Perceptual map of eco-friendly laptop attribute sets

Perceptual maps are graphic representations in which perceptions/preferences of customers are recorded on planes according to varying criteria. They depict perceptual groupings, and thus the proximity of objects, as well as identify their differences according to map measures. By positioning the objects in multidimensional space, using the criteria previously outlined, firms can obtain useful guidelines for their NPD and redesign of existing products (Wind and Robinson, 1972). A vector on the map indicates both the magnitude and the direction of surveyed objects (attributes) in the geographical space. The length of a vector indicates its magnitude. The direction reflects the most preferred direction for an individual (respondent) judge. The axes of the map consist of a special set of vectors that suggest the underlying dimensions that best characterize how customers differentiate among alternatives. Frequently, orthogonal axes (straight lines at right angles) represent the dimensions of the map, though non-orthogonal axes also can be used.

Multidimensional scaling (MDS) techniques provide a set of model algorithms to develop perceptual maps. Using the MDS application from the Statistical Analysis System (SAS) program, we specified a geometric configuration and vector directions for preferences in a common perceptual space. For the map space, we projected eco-friendly laptop attribute locations onto an evaluative criterion vector, indicating the degree of similarity and preference direction of a particular evaluative criterion (see Fig. 2). According to these fundamental characteristics, we can integrate information across maps to determine which evaluative criterion influences each attribute the most.

In Fig. 2, Dimension 1 (the horizontal axis) contains three attribute vectors—energy saving, pollution prevention, and waste recycling—that cluster relatively close together, representing a preference for conservation prevention/recycling. The perceptual map indicates that consumers in the conservation prevention/recycling preference group view battery as the best element, followed by storage device, monitor, CPU, and shell, respectively. The remaining attributes are closer together along the direction of this evaluative criterion vector, indicating that consumers prefer them less and consider them similar in nature; that is, users believe these other attributes simply do not perform effectively on the four evaluative criteria and thus evaluate them negatively.

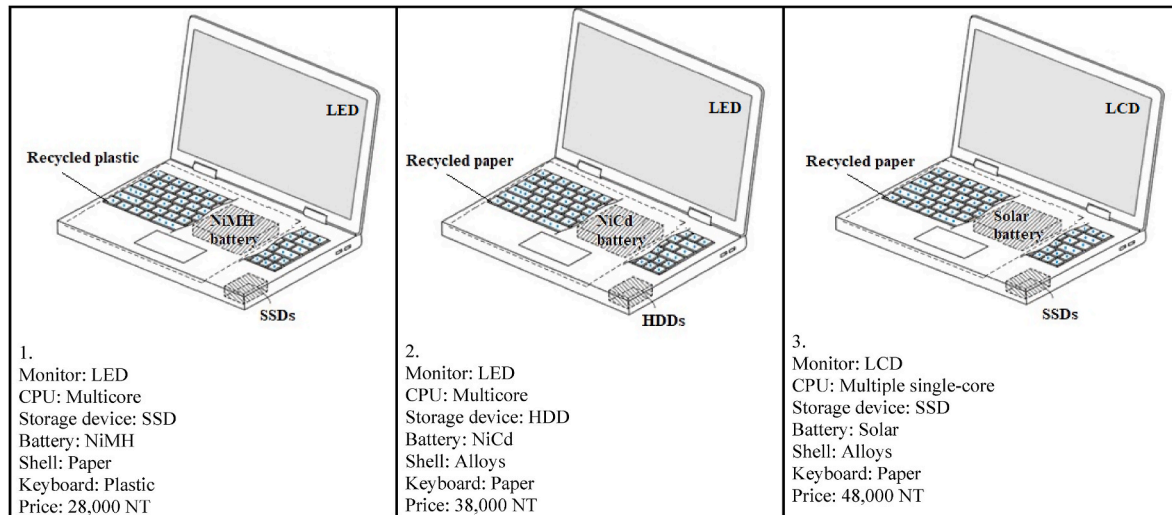


Fig. 1. Example evaluation form describing possible eco-friendly laptop profiles.

Table 2
Respondent characteristics.

Measure	Items	n	Percentage
Gender	Male	154	48.58
	Female	163	51.42
Age(years)	20–29	32	21.77
	30–39	92	29.02
	40–49	124	39.12
	≥50	69	10.09
	≤ Grade 12	157	49.53
Education	> Grade 12	160	50.47
	≤25,000 NT	73	23.03
Income	25,001 NT–45,000 NT	169	53.31
	≥45,001 NT	75	23.66

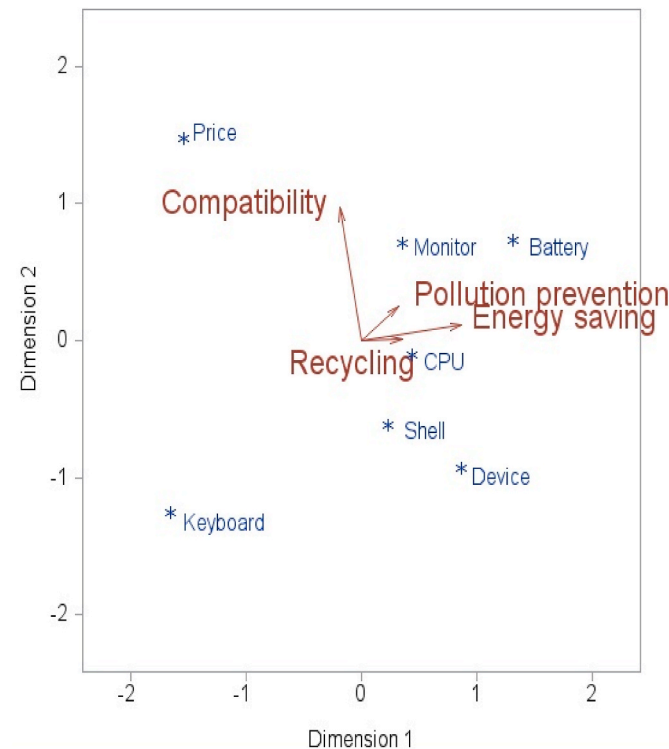


Fig. 2. Perceptual map of eco-friendly laptop attribute sets.

Dimension 2 (the vertical axis) focuses on compatibility; it constitutes the compatibility preference dimension. The perceptual map clearly indicates that if users are members of the compatibility preference group, they view price as the most important element, followed by monitor and battery. Most other attributes are closer together along this criterion vector, indicating that customers prefer them less and consider them similar. In the geographical space, most influential laptop innovation criteria are centralized and far from the other attributes. However, some attributes take sustainable development positions on the map, suggesting opportunities for new products. Finally, the perceptual map indicates that eco-friendly laptop products should feature specific product characteristics to enhance the conservation prevention/recycling and compatibility efforts. We address decisions about these features in our subsequent discussion of the CA method, which reveals that the relative importance attached to various features can be determined from these utilities. This approach also can construct the ratings of experimental profiles that provide choices of all combinations of features and determine which features are the most preferred.

4.3. Conjoint analysis: relative importance of attributes, rank, and part-worth coefficients for optimal eco-friendly laptop products

The CA method is a mathematical technique for determining which features new products should have, by measuring the part-worths obtained from respondents' preference ordering across multi-attribute stimuli, described in terms of product or service features. A part-worth indicates the preference or utility associated with a specific level of an attribute, estimated for each respondent in a conjoint study. Accordingly, utilities are the predicted values from the regression, and part-worth utilities are the regression coefficients; CA reveals the relative importance of each attribute by considering the difference of each one, in terms of the total utility of a product.

As Table 3 shows, the procedure represents the part-worth functions estimated for each level of each attribute, as well as the relative importance of the attributes. The most important attribute is price (24.63%), followed by battery (17.74%), shell (17.53%), and CPU (12.88%). Of intermediate importance are monitor (9.52%), storage device (9.30%), and keyboard (8.40%). As Table 3 indicates, for all respondents, the most preferred combination of eco-friendly laptop consists of median price, solar battery, alloy-based shell, multicore CPU, LED monitor, SSD storage device, and recycled paper keyboard. This combination gives users the highest total part-worth utility value. In addition to assisting in NPD, this finding contributes to the design of personalized laptop products. It describes a simple model of part-worth formation that

Table 3

Estimated attribute-level utility and relative attribute importance for eco-friendly laptop products.

Attributes	Relative importance	Ranking of importance	Levels	Part-worth
Price	24.63%	1	1. 28,000 NT 2. 38,000 NT 3. 48,000 NT	0.0219 0.1901 −0.2120
Battery	17.74%	2	1. Solar battery 2. NiMH battery 3. NiCd battery	0.2046 0.0385 −0.2431
Shell	17.53%	3	1. Recycled paper 2. Recycled plastic 3. Magnesium/Aluminum alloys	−0.0075 −0.0948 0.1023
CPU	12.88%	4	1. Multicore processors 2. Multiple single-core processors	0.0273 −0.0273
Monitor	9.52%	5	1. LED 2. LCD	0.0577 −0.0577
Storage device	9.30%	6	1. HDD 2. SSD	−0.1410 0.1410
Keyboard	8.40%	7	1. Recycled plastic 2. Recycled paper	−0.0418 0.0418

expresses each respondent's utility value for the individual attributes of eco-friendly laptop products. That is, if we have information about a respondent's utilities, we can predict the respondent's rank order of preference for a specific laptop. As Table 4 shows, the product most suitable as a personalized product for Respondent 1 for example is product 26 (28,000 NT, NiCd battery, recycled paper shell, multiple single-core CPU, LCD monitor, SSD storage device, recycled plastic keyboard).

Table 4

Estimated utilities values for one respondent.

Obs.	Utility	Price	Battery	Shell	CPU	Monitor	Device	Keyboard
26	7.92	28,000 NT	NiCd	Paper	Multiple single-core	LCD	SSD	Plastic
32	7.68	48,000 NT	NiCd	Plastic	Multiple single-core	LED	SSD	Plastic
22	7.67	28,000 NT	NiMH	Alloys	Multiple single-core	LCD	SSD	Plastic
9	7.58	28,000 NT	NiCd	Paper	Multiple single-core	LED	HDD	Paper
34	7.55	28,000 NT	Solar	Plastic	Multiple single-core	LED	SSD	Plastic
11	7.52	28,000 NT	NiMH	Paper	Multiple single-core	LED	HDD	Plastic
18	7.47	48,000 NT	Solar	Paper	Multiple single-core	LED	SSD	Paper
31	7.47	48,000 NT	NiMH	Plastic	Multiple single-core	LED	SSD	Paper
29	7.19	38,000 NT	NiCd	Alloys	Multiple single-core	LED	SSD	Plastic
25	7.15	28,000 NT	NiMH	Alloys	Multiple single-core	LCD	HDD	Plastic
8	7.10	48,000 NT	Solar	Alloys	Multiple single-core	LCD	SSD	Paper
35	6.94	48,000 NT	Solar	Paper	Multiple single-core	LED	HDD	Paper
24	6.69	48,000 NT	NiMH	Plastic	Multiple single-core	LCD	HDD	Paper
5	6.46	28,000 NT	NiMH	Paper	Multicore	LED	SSD	Plastic
27	6.45	38,000 NT	NiCd	Paper	Multiple single-core	LCD	HDD	Paper
28	6.42	38,000 NT	Solar	Plastic	Multiple single-core	LCD	SSD	Plastic
12	6.20	38,000 NT	NiMH	Alloys	Multiple single-core	LCD	HDD	Paper
19	6.01	28,000 NT	Solar	Alloys	Multicore	LED	SSD	Paper
20	6.02	28,000 NT	NiCd	Plastic	Multicore	LCD	SSD	Paper
36	5.97	48,000 NT	NiCd	Alloys	Multicore	LCD	SSD	Plastic
10	5.90	38,000 NT	Solar	Plastic	Multiple single-core	LCD	HDD	Plastic
13	5.87	48,000 NT	NiMH	Paper	Multicore	LCD	SSD	Paper
15	5.70	48,000 NT	NiCd	Alloys	Multicore	LED	HDD	Plastic
6	5.57	48,000 NT	NiCd	Plastic	Multicore	LED	HDD	Plastic
17	5.49	28,000 NT	Solar	Alloys	Multicore	LED	HDD	Paper
21	5.48	28,000 NT	NiCd	Plastic	Multicore	LCD	HDD	Paper
33	5.43	48,000 NT	NiMH	Paper	Multicore	LCD	HDD	Plastic
7	5.40	38,000 NT	NiMH	Alloys	Multicore	LED	SSD	Paper
23	5.39	38,000 NT	NiCd	Paper	Multicore	LCD	SSD	Paper
2	5.26	38,000 NT	NiMH	Plastic	Multicore	LED	SSD	Paper
3	5.10	28,000 NT	Solar	Plastic	Multicore	LCD	HDD	Paper
14	5.08	38,000 NT	Solar	Paper	Multicore	LCD	SSD	Plastic
30	5.06	48,000 NT	Solar	Alloys	Multicore	LCD	HDD	Plastic
1	5.01	38,000 NT	NiCd	Alloys	Multicore	LED	HDD	Paper
16	4.82	38,000 NT	NiMH	Plastic	Multicore	LED	HDD	Plastic
4	4.71	38,000 NT	Solar	Paper	Multicore	LED	HDD	Plastic

4.4. Average importance for attributes according to socio-economic profiles

Table 5 shows the relative importance values for eco-friendly laptop attributes according to the socio-economic background of respondents. Price is an important consideration factor for almost all age groups. The age group of 30–39 years assigns higher importance to price and battery, whereas the 40- to 49-year age group gives relatively more importance to compatibility and recycling, reflected by high price and shell utilities. Gender is not an important factor in determining the choice of eco-friendly laptop; the relative importance of all seven attributes is more or less the same among male and female respondents. However, there are differences between lower ($\leq$ Grade 12) and higher ($>$ Grade 12) education levels of respondents in the importance attached to shell and battery. This finding may be correlated with the notion that those with lower education tend to choose shell and battery, while those with higher education seek greater energy savings. Similarly, those in the lower income group ($\leq$ 25,000 NT) perceive price to be an important attribute when choosing an eco-friendly laptop, but those in the higher income group ($\geq$ 45,001 NT) identify battery as more important.

5. Discussion and conclusions

Innovation of green electronics is a core challenge of modern enterprises. This study explores the attribute preferences of user groups as they relate to eco-friendly laptop development; it identifies the relative importance of attributes and investigates whether preferred attributes differ according to types of green electronics. Our analysis of empirical data produces several significant findings with regard to maximizing the user value of price, battery, shell, CPU, monitor, storage device, and keyboard. It reveals that users prefer (1) compatibility with regard to

Table 5

Average importance for attributes according to socio-economic profiles.

Socio-economic variables	Levels	Price (%)	Battery (%)	Shell (%)	CPU (%)	Monitor (%)	Device (%)	Keyboard (%)
Age	20–29 years	25.46	18.71	15.54	13.03	9.55	8.96	8.75
	30–39 years	25.55	17.54	16.82	11.72	9.85	10.41	8.11
	40–49 years	23.46	17.04	19.10	13.50	9.25	9.06	8.59
	≥50 years	24.73	18.90	17.80	13.47	9.53	7.82	7.75
Gender	Male	24.60	17.54	18.26	12.49	9.48	8.96	8.67
	Female	24.66	17.94	16.76	13.29	9.55	9.67	8.13
Education	≤ Grade 12	24.23	16.99	18.66	13.07	9.55	9.09	8.41
	> Grade 12	25.03	18.46	16.42	12.69	9.48	9.52	8.40
Income	≤25,000 NT	26.79	15.49	18.02	11.87	9.44	10.47	7.92
	25,001 NT–45,000 NT	23.71	18.27	17.21	13.84	9.47	9.12	8.38
	≥45,001 NT	24.62	18.71	17.80	11.69	9.69	8.57	8.92

price, energy savings, and pollution prevention; (2) waste recycling with regard to battery; (3) waste recycling with regard to shell; (4) energy savings with regard to CPU; (5) pollution prevention compatibility with regard to monitor; (6) energy savings and recycling with regard to storage device; and (7) usability and recycling with regard to keyboard.

Overall, price is users' most important consideration when they choose eco-friendly laptops. Previous studies (Biswas and Roy, 2014; Lee et al., 2018; Sonnenberg et al., 2014) have supported the importance of price to users' choice of particular eco-friendly products. In our study, prices ranged from 28,000 NT to 38,000 NT; price levels higher than 38,000 NT obtained negative utility values, indicating respondents' reluctance to pay more than that amount for eco-friendly laptops. Price relates highly to compatibility, such that users seem likely to accept prices that are consistent with personal values, past experiences, and needs. Past studies of value-based pricing indicate that when prices match customers' values, preferences, and past experiences, they can increase the company's profitability and sustained success (Dahlstrom, 2010; Danciu, 2018; Hinterhuber, 2008; Reim et al., 2017). As our findings show, users place price ahead of criteria that describe products' environmental friendliness; accordingly, marketers that assume that today's consumers are willing to pay more for eco-efficient appliances may find that providing price compatibility will add consumption value to their products. Such price strategies can be implemented through tactics that give more flexibility to business practices. Moreover, the average degrees of importance for attributes based on age, gender, education, income level, and laptop experience show that price is more important than other attributes for various groups of people. In particular, lower-income (≤25,000 NT) users are more sensitive to changes in the monetary value of the products and more likely to view price as a barrier.

From an environmental perspective, our results show that laptop batteries are important to users in terms of their energy savings, pollution prevention, and waste recycling; this finding is consistent with previous studies (Ansari et al., 2010; Giampieri et al., 2019; Goswami and Kreith, 2016; Luo et al., 2017). Solar-powered batteries in particular are influential determinants of users' eco-friendly laptop choices, so eco-friendly laptop manufacturers might feature solar batteries rather than NiMH batteries without adversely affecting user preferences. The importance of laptop batteries differs according to groups; older respondents (≥50 years) and those with higher levels of education (>Grade 12) and in higher income brackets (≥45,001 NT) ranked them as more important choice factors. Magnesium/aluminum alloys are the most favored type of laptop shell; however, users want all types of shells (recycling paper, alloys) to be recyclable at various quality levels, ranging from recycled paper to alloy materials; the more low-carbon the consumption, the higher the level of sustainable development of human society. Compared with the socio-economic profiles of other characteristics, the sample that favors alloy shells rather than paper shells consists of middle-aged users (40–49 years) in the lower-education bracket (≤Grade 12); this group also favors availability of information about recycling of the material.

Power features continue to play an important role in the design of processors. With regard to CPUs, respondents most prefer multicore processors for saving laptop energy. Because CPUs reduce execution time, balance energy consumption, and minimize total energy used (Coskun et al., 2008; Liao et al., 2017), laptop manufacturers should reap the benefits of multicore processors by designing power-efficient architectures and improving the energy performance of tasks; such initiatives can make users feel sophisticated and invigorated. By appealing to the median income group (25,001 NT–45,000 NT), manufacturers can benefit from the energy resource savings of CPUs. With regard to monitors, users prefer LEDs over LCDs; LED monitors use less energy to provide the same or better service to users, by relying on light-emitting diodes as sources of backlight; LED lighting produces more brilliance and greater light intensity than LCDs and results in annual energy savings of up to 395 TWh—a 60% reduction in electricity consumption (U.S. Department of Energy, 2015). However, product recovery is a good way to satisfy requirements for proper handling and disposal of e-waste; LCD monitors can be manually dismantled to recover critical precious metals, such that their recovery rate reaches more than 90% (Peeters et al., 2013; Savvilotidou et al., 2019; Zhang et al., 2015). Accordingly, laptops with LCD monitors might be well received in electronics markets.

With regard to storage devices, users prefer SSDs over traditional hard drives; they favor higher-interface speeds and lower energy consumption. Because SSDs do not have moving parts for data storage, they are not at risk of any sort of physical damage from movement. This feature improves their start-up times and reduces their power consumption relative to HDDs, which is beneficial when they are installed in battery-powered devices such as laptop computers. Our study shows that lower-income (≤25,000 NT) users regard computational storage devices as notably more important; they assign high utility values to energy-efficient perceptions and practices. It also shows that users prefer typewriter-style keyboards that feature an arrangement of buttons or keys to act as mechanical levers or electronic switches. Such features can provide many benefits of communication and material recycling when plastic or paper materials are recovered and recycled. Ideally, the materials are processed so that they can flow back into the production process.

User participatory design strategies play a key role in green production activity, such that they not only improve the profitability of firms that use co-designed laptop computer products but also remove potentially hazardous materials. Many electronic product developments reflect environmental legislation or recycling and hazardous material demands (e.g., Buekens and Yang, 2014; Patel and Balachandran, 2016; Wong, 2005). However, if these efforts continue without a thorough understanding of various types of knowledge translation, their future development processes are likely to result in a drastic drop in numbers of new electronic construction models. Users can be valuable, accessible fonts of knowledge; they can affect the environmental impacts of products through their choices of environmentally friendly products and practice of environmentally conscious product operations. Not only are

users willing to participate in green activities, such as conserving energy at home, recycling, or driving fuel-efficient cars, but they also are open to participating in electronic product design to express their needs and identify products that fulfill those needs.

Our study shows that eco-friendly laptop designers should emphasize the characteristics of price, battery, shell, CPU, monitor, storage device, and keyboard to maximize user value. Development trends change quickly in the electronics industry; developments in energy-efficient laptop components continue to emerge and provide new possibilities for satisfying users' perceptual demands. Our findings suggest that green laptop design is proceeding in the right direction by emphasizing the creation of environmental attributes according to user collaboration. This increase in environmental consciousness has had a positive effect on consumer behaviors and helped expand the electric and electronic equipment market at a remarkable rate.

References

- Aguilera-Caracuel, J., Ortiz-de-Mandojana, N., 2013. Green innovation and financial performance: an institutional approach. *Organ Environ* 26 (4), 365–385. <https://doi.org/10.1177/1086026613507931>.
- Al Assaf, M.M., Jiang, X., Abid, M.R., Qin, X., 2013. Eco-storage: a hybrid storage system with energy-efficient informed prefetching. *J Signal Process Syst* 72 (3), 165–180. <https://doi.org/10.1007/s11265-013-0784-9>.
- Alriksson, S., Öberg, T., 2008. Conjoint analysis for environmental evaluation: a review of methods and applications. *Environ Sci Pollut Res* 15 (3), 244–257. <https://doi.org/10.1006/espr2008.02.479>.
- Andersson, G., 2004. Power electronics solutions for sustainability. In: Proceedings of the 2004 IEEE Power Engineering Society General Meeting. IEEE, pp. 2304–2308. <http://ieeexplore.ieee.org/document/1373297>.
- Ansari, N.L., Ashraf, M.M., Malik, B.T., Grunfeld, H., 2010. Green IT awareness and practices: results from a field study on mobile phone related e-waste in Bangladesh. In: Proceedings of the 2010 IEEE International Symposium on Technology and Society. IEEE, pp. 375–383. <http://ieeexplore.ieee.org/document/5514618>.
- Awasthi, A.K., Li, J., 2017. Management of electrical and electronic waste: a comparative evaluation of China and India. *Renew Sust Energ Rev* 76, 434–447. <https://doi.org/10.1016/j.rser.2017.02.067>.
- Ayvaz, B., Bolat, B., Aydin, N., 2015. Stochastic reverse logistics network design for waste of electrical and electronic equipment. *Resour Conserv Recycl* 104 (Part B), 391–404. <https://doi.org/10.1016/j.resconrec.2015.07.006>.
- Baldé, C.P., Wang, F., Kuehr, R., Huisman, J., 2015. The Global e-Waste Monitor– 2014. United Nations University, Bonn.
- Berchicci, L., Bodewes, W., 2005. Bridging environmental issues with new product development. *Bus Strateg Environ* 14 (5), 272–285. <https://doi.org/10.1002/bse.488>.
- Bhatia, M., Jain, A., 2013. Green marketing: a study of consumer perception and preferences in India. *Electron Green J* 36, 1–19. <https://doi.org/10.5070/G313618392>.
- Biswas, A., Roy, M., 2014. Green products: an exploratory study on the consumer behaviour in emerging economies of the East. *J Clean Prod* 87, 463–468. <https://doi.org/10.1016/j.jclepro.2014.09.075>.
- Blome, C., Hollos, D., Paulraj, A., 2014. Green procurement and green supplier development: antecedents and effects on supplier performance. *Int J Prod Res* 52 (1), 32–49. <https://doi.org/10.1080/00207543.2013.825748>.
- Boger, P., Dusthimer, D., 2013. IT Essentials: PC Hardware and Software Companion Guide, fifth ed. Cisco Press, Indianapolis.
- Buekens, A., Yang, J., 2014. Recycling of WEEE plastics: a review. *J Mater Cycles Waste Manag* 16 (3), 415–434. <https://doi.org/10.1007/s10163-014-0241-2>.
- Chan, H.K., Yee, R.W.Y., Dai, J., Lim, M.K., 2016. The moderating effect of environmental dynamism on green product innovation and performance. *Int J Prod Econ* 181 (Part B), 384–391. <https://doi.org/10.1016/j.ijpe.2015.12.006>.
- Chang, C.H., 2019. Do green motives influence green product innovation? the mediating role of green value co-creation. *Corp Soc Responsib Environ Manag* 26 (2), 330–340. <https://doi.org/10.1002/csr.1685>.
- Chang, C.H., Chen, Y.S., 2013. Green organizational identity and green innovation. *Manag Decis* 51 (5), 1056–1070. <https://doi.org/10.1108/MD-09-2011-0314>.
- Chayet, S., Hopp, W.J., Xu, X., 2004. The marketing-operations interface. In: Simchi-Levi, D., Wu, S.D., Shen, Z.J. (Eds.), *Handbook of Quantitative Supply Chain Analysis: Modeling in the E-Business Era*. Springer, New York, pp. 295–334.
- Chen, R.J.C., 2014. An integrated sustainable business and development system: thoughts and opinions. *Sustainability* 6 (10), 6862–6871. <https://doi.org/10.3390/su6106862>.
- Chen, P.C., Hung, S.W., 2014. Collaborative green innovation in emerging countries: a social capital perspective. *Int J Oper Prod Manage* 34 (3), 347–363. <https://doi.org/10.1108/IJOPM-06-2012-0222>.
- Chen, Y.S., Lai, S.B., Wen, C.T., 2006. The influence of green innovation performance on corporate advantage in Taiwan. *J Bus Ethics* 67 (4), 331–339. <https://doi.org/10.1007/s10551-006-9025-5>.
- Chen, Y.S., Chang, C.H., Wu, F.S., 2012. Origins of green innovations: the differences between proactive and reactive green innovations. *Manag Decis* 50 (3), 368–398. <https://doi.org/10.1108/00251741211216197>.
- Chesbrough, H.W., 2006. Open innovation: a new paradigm for understanding industrial innovation. In: Chesbrough, H.W., Vanhaverbeke, W., West, J. (Eds.), *Open Innovation: Researching a New Paradigm*. Oxford University Press, Oxford, pp. 1–12.
- Cho, Y., Koo, Y., Huh, S.Y., Lee, M., 2015. Evaluation of a consumer incentive program for an energy-efficient product in South Korea. *Energy Effic* 8 (4), 745–757. <https://doi.org/10.1007/s12053-014-9319-x>.
- Ciocci, R., Pecht, M., 2006. Impact of environmental regulations on green electronics manufacture. *Microelectron. Int* 23 (2), 45–50. <https://doi.org/10.1108/13565360610659716>.
- Coskun, A.K., Rosing, T.S., Whisnant, K.A., Gross, K.C., 2008. Static and dynamic temperature-aware scheduling for multiprocessor SoCs. *IEEE Trans Very Large Scale Integr Syst* 16 (9), 1127–1140. <https://doi.org/10.1109/tvlsi.2008.2000726>.
- Curran, M.A., 2013. Life cycle assessment: a review of the methodology and its application to sustainability. *Curr Opin Chem Eng* 2 (3), 273–277. <https://doi.org/10.1016/j.coche.2013.02.002>.
- Dahlstrom, R., 2010. *Green Marketing Management*. South-Western Cengage Learning, Ohio.
- Danciuc, V., 2018. The Green marketing focus changing: from ecological to sustainable marketing (II). *Rom Econ J* 21 (67), 145–193.
- Dangelico, R.M., 2016. Green product innovation: where we are and where we are going. *Bus Strateg Environ* 25 (8), 560–576. <https://doi.org/10.1002/bse.1886>.
- Dangelico, R.M., Pujari, D., 2010. Mainstreaming green product innovation: why and how companies integrate environmental sustainability. *J Bus Ethics* 95 (3), 471–486. <https://doi.org/10.1007/s10551-010-0434-0>.
- DeMendonça, M., Baxter, T.E., 2001. Design for the environment (DFE)—an approach to achieve the ISO 14000 international standardization. *Environ Manag Health* 12 (1), 51–56. <https://doi.org/10.1108/09566160110381922>.
- IF Design, 2021. Paper PP Alloy/Partially recycled material for notebooks. [online]. Available at: <https://ifworlddesignguide.com/entry/80998-paper-pp-alloy>. (Accessed 1 August 2021).
- Dexigner, 2011. Three PEGA D&E Products Win IF Design Award for 2011 [online]. Available at: <https://www.dexigner.com/news/22252>. (Accessed 1 August 2021).
- Díaz-García, C., González-Moreno, A., Sáez-Martínez, F.J., 2015. Eco-innovation: insights from a literature review. *Innov-Manag Policy Pract* 17 (1), 6–23. <https://doi.org/10.1080/14479338.2015.1011060>.
- DisplaySearch, 2011. LED & CCFL Backlight Cost Report second quarter 2011.
- Driessen, P., Hillebrand, B., 2002. Adoption and diffusion of green innovations. In: Nelissen, W., Bartels, G. (Eds.), *Marketing for Sustainability: towards Transactional Policy-Making*. IOS Press Inc, Amsterdam, pp. 343–356.
- Dumitrescu, L., Maxineasa, S.G., Simion, I.M., Taranu, N., Andrei, R., Gavrilescu, M., 2014. Evaluation of the environmental impact of road pavements from a life cycle perspective. *Environ Eng Manag J* 13 (2), 449–455. <https://doi.org/10.30638/eemj.2014.049>.
- Ellram, L.M., Tate, W., Carter, C.R., 2008. Applying 3DCE to environmentally responsible manufacturing practices. *J Clean Prod* 16 (15), 1620–1631. <https://doi.org/10.1016/j.jclepro.2008.04.017>.
- Esmaili, M., Fazeli, S.F., 2015. Surveying of importance of green marketing compared purchase budget and preferred brand when buying by AHP method. *Med J Soc Sci* 6 (4), 388–394. <https://doi.org/10.5901/mjss.2015.v6n4p388>.
- EC (European Commission), 2000. Proposal for a Directive of the European Parliament and of the Council on Waste, Electrical and Electronic Equipment, 2000. COM, Brussels, 347 final.
- Farber, S., Griner, B., 2000. Using conjoint analysis to value ecosystem change. *Environ Sci Technol* 34 (8), 1407–1412. <https://doi.org/10.1021/es990727r>.
- Finnveden, G., Hauschild, M.Z., Ekvall, T., Guinee, J., Heijungs, R., Hellweg, S., Koehler, A., Pennington, D., Suh, S., 2009. Recent developments in life cycle assessment. *J Environ Manage* 91 (1), 1–21. <https://doi.org/10.1016/j.jenvman.2009.06.018>.
- Fitzgerald, E., Wankerl, A., Schramm, C., 2011. *Inside Real Innovation: How the Right Approach Can Move Ideas from R&D to Market — and Get the Economy Moving*. World Scientific Publishing, Singapore.
- Fletcher, K.T., Goggin, P.A., 2001. The dominant stances on ecodesign: a critique. *Design Issue* 17 (3), 15–25. <https://doi.org/10.1162/074793601750357150>.
- Freeman, C., 1982. *The Economics of Industrial Innovation*. MIT Press, Cambridge, MA.
- Gao, M., Shih, C.C., Pan, S.Y., Chueh, C.C., Chen, W.C., 2018. Advances and challenges of green materials for electronics and energy storage applications: from design to end-of-life recovery. *J Mater Chem A* 6 (42), 20546–20563. <https://doi.org/10.1039/C8TA07246A>.
- Geffen, C.A., Rothenberg, S., 2000. Suppliers and environmental innovation: the automotive paint process. *Int J Oper Prod Manage* 20 (2), 166–186. <https://doi.org/10.1108/01443570010304242>.
- Giamperli, A., Ling-Chin, J., Taylor, W., Smallbone, A., Roskilly, A.P., 2019. Moving towards low-carbon manufacturing in the UK automotive industry. *Energy Procedia* 158, 3381–3386. <https://doi.org/10.1016/j.egypro.2019.01.946>.
- Goswami, D.Y., Kreith, F., 2016. *Energy Efficiency and Renewable Energy Handbook*, second ed. CRC Press, Boca Raton.
- Green, P.E., Krieger, A.M., Wind, Y., 2001. Thirty years of conjoint analysis: reflections and prospects. *Interfaces* 31 (3), S56–S73. <https://doi.org/10.1287/inte.31.4.56.9676>.
- Gronow, J., 2009. Fads, fashions and 'real' innovations: novelties and social change. In: Shove, E., Trentmann, F., Wilk, R. (Eds.), *Time, Consumption and Everyday Life: Practice, Materiality and Culture*. Berg, Oxford, pp. 129–142.
- Gu, F., Ma, B., Guo, J., Summers, P.A., Hall, P., 2017. Internet of things and big data as potential solutions to the problems in waste electrical and electronic equipment management: an exploratory study. *Waste Manage* 68, 434–448. <https://doi.org/10.1016/j.wasman.2017.07.037>.

- Guarte, J.M., Barrios, E.B., 2006. Estimation under purposive sampling. *Commun Stat – Simul C 35* (2), 277–284. <https://doi.org/10.1080/03610910600591610>.
- Gustafsson, A., Herrmann, A., Huber, F., 2003. *Conjoint Measurement: Methods and Applications*, third ed. Springer, Berlin.
- Hansen, E.G., Grosse-Dunker, F., Reichwald, R., 2009. Sustainability innovation cube—a framework to evaluate sustainability-oriented innovations. *Int J Innov Manag* 13 (4), 683–713. <https://doi.org/10.1142/S1363919609002479>.
- Hensher, D.A., 2006. How do respondents process stated choice experiments? Attribute consideration under varying information load. *J Appl Econom* 21 (6), 861–878. <https://doi.org/10.1002/jae.877>.
- Hensher, D.A., Rose, J.M., Greene, W.H., 2012. Inferring attribute non-attendance from stated choice data: an application for willingness to pay estimates and a warning for stated choice experiment design. *Transportation* 39 (2), 235–245. <https://doi.org/10.1007/s11116-011-9347-8>.
- Hinterhuber, A., 2008. Customer value-based pricing strategies: why companies resist. *J Bus Strateg* 29 (4), 41–50. <https://doi.org/10.1108/02756660810887079>.
- Hsu, C.C., Tan, K.C., Zailani, S.H.M., Jayaraman, V., 2013. Supply chain drivers that foster the development of green initiatives in an emerging economy. *Int J Oper Prod Manage* 33 (6), 656–688. <https://doi.org/10.1108/IJOPM-10-2011-0401>.
- Huang, Y.C., Wu, Y.C.J., 2010. The effects of organizational factors on green new product success: evidence from high-tech industries in Taiwan. *Manag Decis* 48 (10), 1539–1567. <https://doi.org/10.1108/00251741011090324>.
- Jabbour, C.J.C., Jugend, D., de Sousa Jabbour, A.B.L., Gunasekaran, A., Latan, H., 2015. Green product development and performance of Brazilian firms: measuring the role of human and technical aspects. *J Clean Prod* 87, 442–451. <https://doi.org/10.1016/j.jclepro.2014.09.036>.
- Jannat, U.K., 2016. Green software engineering adaption in requirement elicitation process. *Int J Sci Technol Res* 5 (8), 94–98.
- Jay, S., Jones, C., Slinn, P., Wood, C., 2007. Environmental impact assessment: retrospect and prospect. *Environ Impact Assess Rev* 27 (4), 287–300. <https://doi.org/10.1016/j.eiar.2006.12.001>.
- Joshi, Y., Rahman, Z., 2016. Predictors of young consumer's green purchase behaviour. *Manag Environ Qual* 27 (4), 452–472. <https://doi.org/10.1108/MEQ-05-2015-0091>.
- Kadirov, D., 2015. Private labels ain't bona fide! Perceived authenticity and willingness to pay a price premium for national brands over private labels. *J Market Manag* 31 (17–18), 1773–1798. <https://doi.org/10.1080/0267257X.2015.1031265>.
- Kaewunruen, S., Rungskunroch, P., Jennings, D.V., 2019. A through-life evaluation of end-of-life rolling stocks considering asset recycling, an energy recovering, and financial benefit. *J Clean Prod* 212, 1008–1024. <https://doi.org/10.1016/j.jclepro.2018.11.271>.
- Kumar, A.P., Kannegala, S.S., 2012. Green devices and hardware. In: Murugesan, S., Gangadharan, G.R. (Eds.), *Harnessing Green IT: Principles and Practices*. Wiley, New York, pp. 23–38.
- Laville, S., Taylor, M., 2017. A million bottles a minute: world's plastic binge 'as dangerous as climate change' [online]. Available at: <https://www.theguardian.com/environment/2017/jun/28/a-million-a-minute-worlds-plastic-bottle-binge-as-dangerous-as-climate-change>. (Accessed 7 August 2019).
- Lawson, S.R., Roggenbuck, J.W., Hal, T.E., Moldovanyi, A., 2006. A conjoint analysis of preference heterogeneity among day and overnight visitors to the Okefenokee wilderness. *J Leis Res* 38 (4), 575–600. <https://doi.org/10.1080/00222216.2006.11950093>.
- Lee, J., Bhatt, S., Suri, R., 2018. When consumers penalize not so green products. *Psychol Mark* 35 (1), 36–46. <https://doi.org/10.1002/mar.21069>.
- Li, G., Lim, M., Wang, Z., 2019. Stakeholders, green manufacturing, and practice performance: empirical evidence from Chinese fashion businesses. *Ann Oper Res* 290, 961–982. <https://doi.org/10.1007/s10479-019-03157-7>.
- Liao, R., 2019. How to produce novelty? creating, borrowing, modifying, repeating and forgetting: the process of contemporary fashion aesthetics. *AM J Art Media Stud* 19, 101–107. <https://doi.org/10.25038/am.v019i.310>.
- Liao, Z., 2019. Social capital and firms' environmental innovations: the moderating role of environmental scanning. *Bus Strateg Environ* 27 (8), 1493–1501. <https://doi.org/10.1002/bse.2207>.
- Liao, S.L., Liu, B.-X., Cheng, C.T., Li, Z.F., Wu, X.Y., 2017. Long-term generation scheduling of hydropower system using multi-core parallelization of particle swarm optimization. *Water Resour Manag* 31 (3), 2791–2807. <https://doi.org/10.1007/s11269-017-1662-1>.
- Lin, Y.H., Chen, Y.S., 2016. Determinants of green competitive advantage: the roles of green knowledge sharing, green dynamic capabilities, and green service innovation. *Qual Quant* 51, 1663–1685. <https://doi.org/10.1007/s11135-016-0358-6>.
- Liu, S., Webb, H., Zhou, H.W., 2019. Green innovation in challenging institutional contexts: a case study on Landsea's environmental projects. *Ekoloji* 107, 3327–3338.
- Luo, B., Ye, D., Wang, L., 2017. Recent progress on integrated energy conversion and storage systems. *Adv Sci* 4 (9), 1700104. <https://doi.org/10.1002/adv.201700104>.
- Ma, Y., Hou, G., Xin, B., 2017. Green process innovation and innovation benefit: the mediating effect of firm image. *Sustainability* 9, 1778. <https://doi.org/10.3390/su9101778>.
- Mahesri, A., Vardhan, V., 2005. Power consumption breakdown on a modern laptop. In: Falsafi, B., VijayKumar, T.N. (Eds.), *PACS 2004*. LNCS, vol. 3471. Springer, Heidelberg, pp. 165–180.
- Mardani, A., Streimikiene, D., Zavadskas, E.K., Cavallaro, F., Nilashi, M., Jusoh, A., Zare, H., 2017. Application of structural equation modeling (SEM) to solve environmental sustainability problems: a comprehensive review and meta-analysis. *Sustainability* 9 (10), 1814. <https://doi.org/10.3390/su9101814>.
- Mauro, E.D., Hebrard, E., Boulahia, Y., Rolandi, M., Santato, C., 2019. On the interfaces between organic bio-sourced materials and metals for sustainable electronics: the eumelanin case. *Jpn J Appl Phys* 58 (5), 051014. <https://doi.org/10.7567/1347-4065/ab1061>.
- Mueller, J., Griese, H., Schischke, K., Stobbel, L., 2005. Transition to lead free soldering—a great change for a better understanding of materials and processes and green electronics. In: *Proceedings of the 2005 International Conference on Asian Green Electronics*. IEEE, pp. 33–36. <http://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=1452312&isnu mber=31035>.
- Nunnally, J., 1978. *Psychometric Theory*. McGraw-Hill, New York.
- Park, W.Y., Phadke, A., Shah, N., 2013. Efficiency improvement opportunities in personal computer monitors: implications for market transformation programs. *Energy Effic* 6 (3), 545–569. <https://doi.org/10.1007/s12053-013-9191-0>.
- Patel, B., Balachandran, V., 2016. E-waste: health hazards and legal environment in India. *Int J Adv Res Manag Soc Sci* 5 (12), 119–129.
- Patton, M.Q., 2002. *Qualitative Research and Evaluation Methods*. Sage, Thousand Oaks, CA.
- Peeters, J.R., Vanegas, P., Duflo, J.R., Mizuno, T., Fukushima, S., Umeda, Y., 2013. Effects of boundary conditions on the end-of-life treatment of LCD TVs. *CIRP Ann-Manuf Technol* 62 (1), 35–38. <https://doi.org/10.1016/j.cirp.2013.03.092>.
- Porter, M.E., van der Linde, C., 1995. Green and competitive: ending the stalemate. *Harv Bus Rev* 73 (5), 120–134.
- Rao, V.R., 2010. Conjoint analysis. In: Sheth, J., Malhotra, N. (Eds.), *Wiley International Encyclopedia of Marketing*. John Wiley & Sons, Chichester, UK. <https://doi.org/10.1002/9781444316568.wiem02019>.
- Reim, W., Lenka, S., Frishammar, J., Parida, V., 2017. Implementing sustainable product—service systems utilizing business model activities. *Procedia CIRP* 64, 61–66. <https://doi.org/10.1016/j.procir.2017.03.130>.
- Roberts, E.B., 1988. What we've learned: managing invention and innovation. *Res-Technol Manag* 31 (1), 11–29. <https://doi.org/10.1080/08956308.1988.11670497>.
- Robinson, D.E., 1958. Fashion theory and product design. *Harvard Bus Rev* 36 (6), 126–138.
- Rogers, E.M., 2003. *Diffusion of Innovations*, fifth ed. Free Press, New York.
- Rosen, S., 1974. Hedonic prices and implicit markets: product differentiation in pure competition. *J Polit Econ* 82 (1), 34–55. <https://doi.org/10.1086/260169>.
- Sancha, C., Wong, C.W., Thomsen, C.G., 2016. Buyer-supplier relationships on environmental issues: a contingency perspective. *J Clean Prod* 112 (3), 1849–1860. <https://doi.org/10.1016/j.jclepro.2014.09.026>.
- Savvilitidou, V., Kousaiti, A., Batinic, B., Vaccari, M., Kastanaki, E., Karagianni, K., Gidarokos, E., 2019. Evaluation and comparison of pre-treatment techniques for recovering indium from discarded liquid crystal displays. *Waste Manage* 87, 51–61. <https://doi.org/10.1016/j.wasman.2019.01.029>.
- Schiederig, T., Tietze, F., Herstatt, C., 2012. Green innovation in technology and innovation management—an exploratory literature review. *R D Manage* 42 (2), 180–192. <https://doi.org/10.1111/j.1467-9310.2011.00672.x>.
- Sharma, S., Henriques, I., 2005. Stakeholder influences on sustainability practices in the Canadian forest products industry. *Strateg Manag J* 26 (2), 159–180. <https://doi.org/10.1002/smj.439>.
- Shimada, T., Van Wassenhove, L.N., 2019. Closed-Loop supply chain activities in Japanese home appliance/personal computer manufacturers: a case study. *Int J Prod Econ* 212, 259–265. <https://doi.org/10.1016/j.ijpe.2016.11.010>.
- Simmel, G., 1957. Fashion. *Am J of Sociol* 62 (6), 541–558.
- Sonnenberg, N.C., Erasmus, A.C., Schreuder, A., 2014. Consumers' preferences for eco-friendly appliances in an emerging market context. *Int J Consum Stud* 38 (5), 559–569. <https://doi.org/10.1111/ijcs.12120>.
- Sun, Z.H.I., Xiao, Y., Sietsma, J., Agterhuis, H., Visser, G., Yang, Y., 2015. Selective copper recovery from complex mixtures of end-of-life electronic products with ammonia-based solution. *Hydrometallurgy* 152, 91–99. <https://doi.org/10.1016/j.hydromet.2014.12.013>.
- Tabassum, F., Bari, Q.H., Rahman, M.M., Mahmud, M.M., Raj, A.R., 2017. Printing press solid waste recycling in Dhaka city. In: *Proceedings of the 5th International Conference on Solid Waste Management in South Asian Countries*, pp. 25–27. WasteSafe. https://www.researchgate.net/publication/314256921_PRINTING_PRESS_SOLID_WASTE_RECYCLING_IN_DHAKA_CITY.
- Thompson, S., 2019. Electronic waste and its regulation. In: Themelis, N., Bourtsalas, A. (Eds.), *Recovery of Materials and Energy from Urban Wastes*. Encyclopedia of Sustainability Science and Technology Series. Springer, New York, pp. 119–124.
- Thongmak, M., 2016. Youths' green information and communications technology acceptance and implications for the innovation decision process. *Electron Green J* 1 (39), 1–25.
- Tsai, K.H., Liao, Y.C., 2017. Innovation capacity and the implementation of eco-innovation: toward a contingency perspective. *Bus Strateg Environ* 26 (7), 1000–1013. <https://doi.org/10.1002/bse.1963>.
- U.S. Department of Energy, 2015. Solid-state lighting R&D plan [online]. Available at: https://www.energy.gov/sites/prod/files/2015/06/f22/ssl_rd-plan_may2015_0.pdf. (Accessed 15 August 2019).
- Vermeulen, J., Verwey, R., Hochstenbach, L., Weegen, S.V., Man, Y.P., Witte, L.D., 2014. Experiences of multidisciplinary development team members during user-centered design of telecare products and services: a qualitative study. *J Med Internet Res* 16 (5), e124. <https://doi.org/10.2196/jmir.3195>.
- Vinodh, S., Rathod, G., 2010. Integration of ECQFD and LCA for sustainable product design. *J Clean Prod* 18 (8), 833–842. <https://doi.org/10.1016/j.jclepro.2009.12.024>.
- Virtanen, Y., Nilsson, S., 1993. *Environmental Impacts of Waste Paper Recycling*. Earthscan Publications Ltd, London.

- Vyas, N., Woodside, A.G., 1984. An inductive model of industrial supplier choice Processes. *J Market* 48 (1), 30–45. <https://doi.org/10.1177/002224298404800103>.
- Wang, X.V., Wang, L., 2017. A cloud-based production system for information and service integration: an internet of things case study on waste electronics. *Enterp Inf Syst* 11 (7), 952–968. <https://doi.org/10.1080/17517575.2016.1215539>.
- Wind, Y., Robinson, P.J., 1972. Product positioning: an application of multidimensional scaling. In: Haley, R.I. (Ed.), *Attitude Research in Transition*. American Marketing Association, Chicago, pp. 155–175.
- Wong, S.F., 2005. Obliging institutions and industry evolution: a comparative study of the German and UK wind energy industries. *Ind Innov* 12 (1), 117–145. <https://doi.org/10.1080/1366271042000339085>.
- Zailani, S., Govindan, K., Iranmanesh, M., Shaharudin, M.R., Chong, Y.S., 2015. Green innovation adoption in automotive supply chain: the Malaysian case. *J Clean Prod* 108 (Part A), 1115–1122. <https://doi.org/10.1016/j.jclepro.2015.06.039>.
- Zhang, K., Wu, Y., Wang, W., Li, B., Zhang, Y., Zuo, T., 2015. Recycling indium from waste LCDs: a review. *Resour Conserv Recycl* 104 (Part A), 276–290. <https://doi.org/10.1016/j.resconrec.2015.07.015>.